

FitBuddy – AI Fitness Plan Generator

Module 8: Project Demonstration & Live Execution Guide

MODULE
8. Project Demonstration

CORE MODEL
Google Gemini

DEMO SCOPE
End-to-End Walkthrough

REPOSITORY
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Demo 1 Application Startup & Interface Initialization

The demonstration begins by launching the FitBuddy interface and connecting securely to the Gemini API endpoint using standard environment configuration keys.

```
$ fitbuddy-app launch
[INFO] Initializing FitBuddy Engine...
[INFO] Loading Google Gemini API Model Driver... Connected (Gemini Model Ready)
[INFO] Server listening on http://localhost:3000
```

Demo 2 User Profile Input Workflow

The user inputs their personalized parameters into the FitBuddy questionnaire wizard. These variables drive the custom prompt built for the Gemini AI generator.

Input Field	Sample Demo Parameter	Purpose in Gemini Prompt
Fitness Goal	Muscle Hypertrophy & Strength	Sets exercise repetition ranges & split type
Experience Level	Intermediate	Controls exercise complexity and volume
Weekly Frequency	4 Days / Week (Upper / Lower Split)	Structures 7-day schedule distribution
Available Equipment	Dumbbells, Pull-up Bar, Resistance Bands	Filters out unavailable gym machinery
Dietary Preference	High Protein Vegetarian	Shapes macronutrient & meal suggestion output

Demo 3 Real-time AI Generation & Processing

Upon clicking "**Generate Plan**", the application formats the request into a structured JSON schema prompt sent to the Gemini model.

Prompt Payload Construction

User fields are wrapped in system safety instructions to prevent unsafe caloric targets or high-risk movement combinations.

Streaming & Parsing

Gemini streams back raw structured JSON, which FitBuddy's renderer validates against predefined TypeScript interfaces.

Demo 4

Generated Output Display & Visual Plan

Below is a sample of the formatted weekly routine output generated by Gemini during the live project demonstration.

Day / Split	Primary Focus	Sample Generated Exercises	Target Sets & Reps
Day 1: Upper Body A	Chest, Back & Arms	Dumbbell Bench Press, Inverted Rows	4 Sets × 8–10 Reps
Day 2: Lower Body A	Quads & Core	Goblet Squats, Bulgarian Split Squats	3 Sets × 10–12 Reps
Day 3: Active Rest	Recovery & Mobility	Light Walking, Banded Shoulder Mobility	20–30 Minutes
Day 4: Upper Body B	Shoulders & Pulls	Pull-ups, Dumbbell Overhead Press	4 Sets × 8–10 Reps
Day 5: Lower Body B	Posterior Chain	Dumbbell Romanian Deadlifts, Calf Raises	3 Sets × 12–15 Reps

Demo 5

Plan Export & Customization Features

The demonstration highlights key interactive features available after the AI plan has been generated.

PDF Export Feature

Users can export their complete 7-day schedule and meal blueprint directly into a formatted PDF document with a single click.

Dynamic Plan Refinement

If an exercise is unsuitable, the user can request Gemini to substitute a specific movement without regenerating the entire week.

Demo Summary

Key Takeaways

Demonstration Conclusion

The demonstration confirms that **FitBuddy** successfully leverages Google Gemini models to deliver fast, highly customized, and structurally reliable fitness plans through a user-friendly interface.

